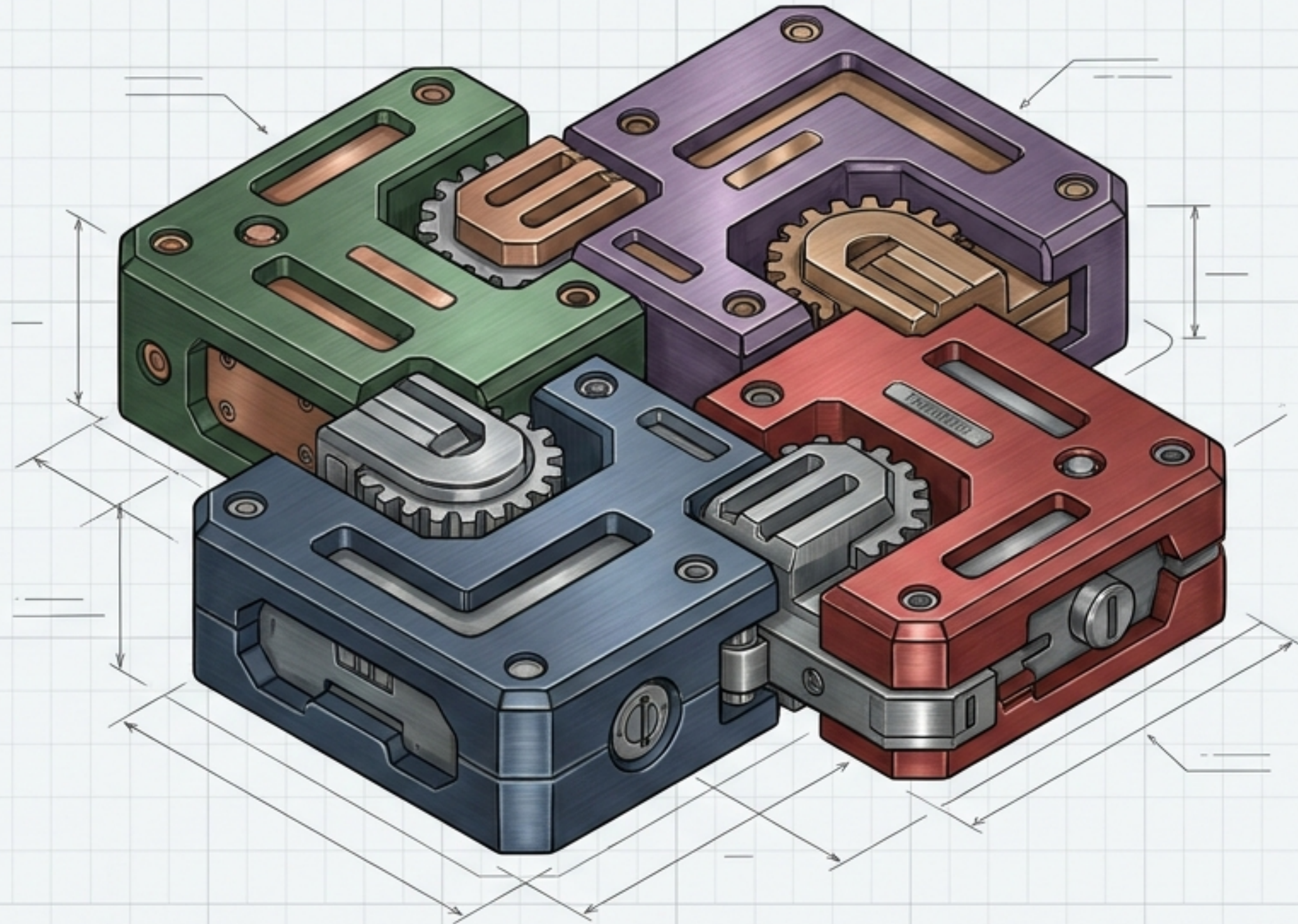


Architecting the 'Two Brains' AI Enterprise

A Strategic Blueprint for Combining Predictive ML and Generative AI in Production



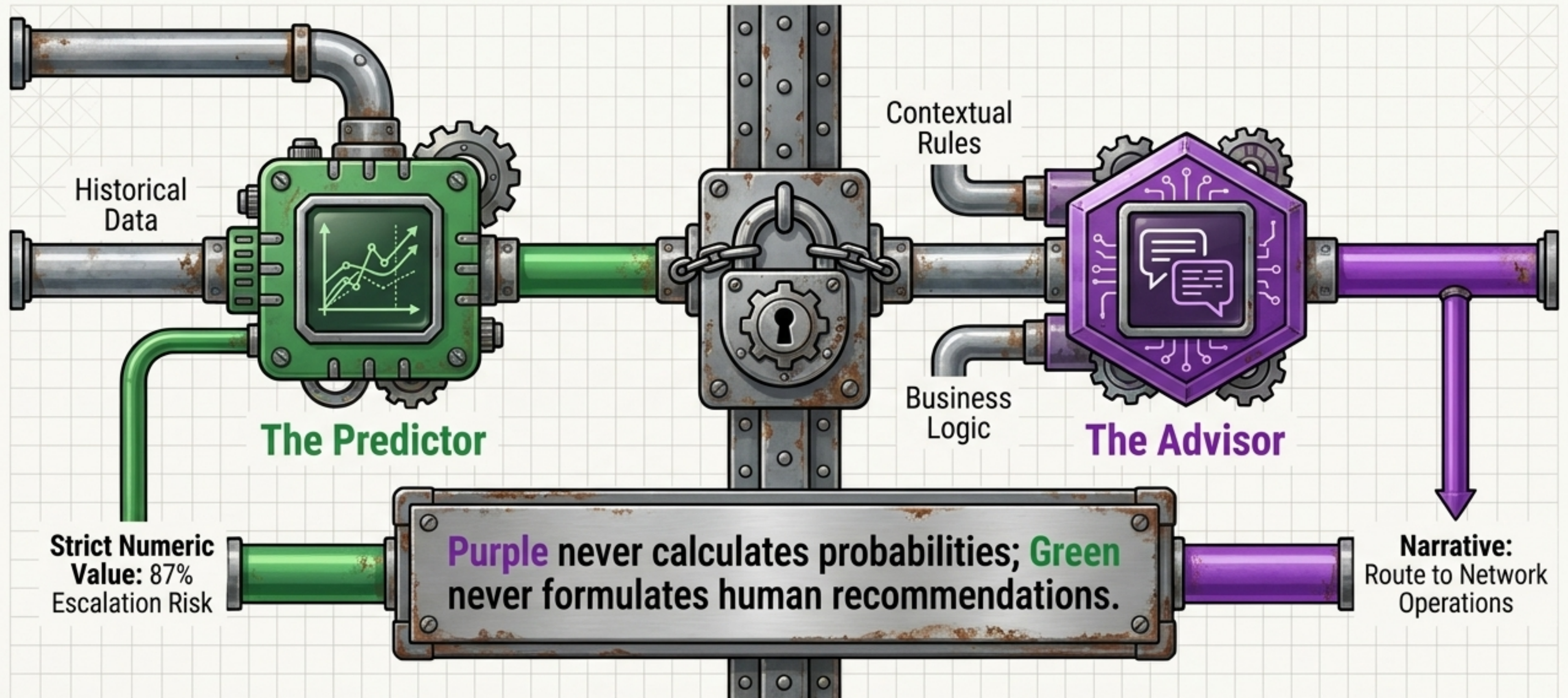
The Fallacy of Singular AI

The most useful mental model is not two kinds of AI glued together. It is a decision system with strictly separated responsibilities. Neither capability can safely solve enterprise problems alone.

	Predictive ML	Generative AI	Optimization & Rules
Superpower	Learning repeatable relationships from historical data.	Synthesizing heterogeneous context and conversational interaction.	Enforcing compliance, budgets, capacities, and deterministic invariants.
Fatal Flaw	Correlation is not causation; produces a raw score without operational context.	Confabulation; omits constraints; persuasive but unsupported reasoning.	Brittle for ambiguous or novel situations.
Output	Probability, forecast interval, anomaly score.	Recommendation narrative, rationale, structured draft.	Optimal action set; Allow/Deny.

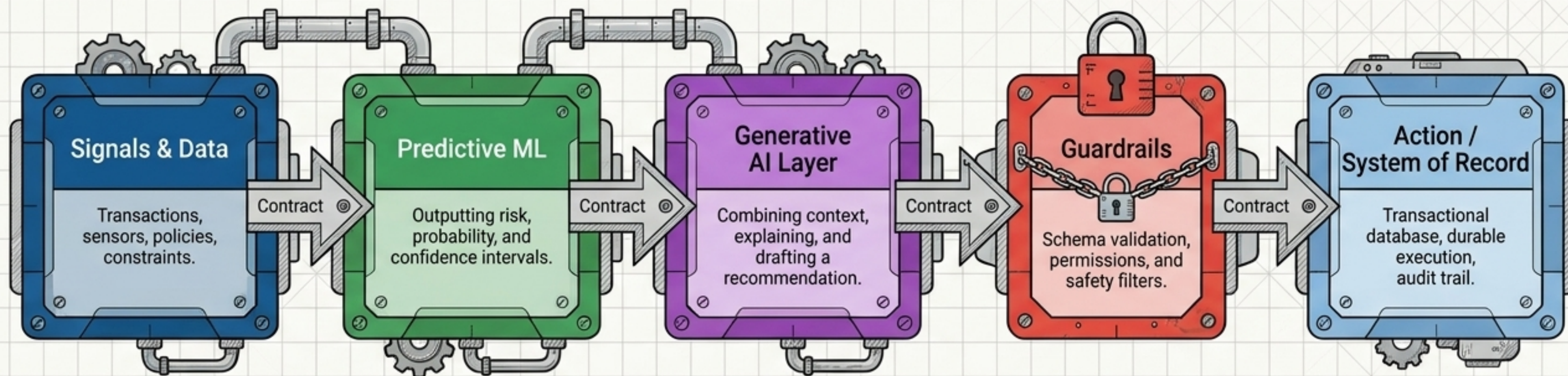
The Immutable Boundary: Separating Estimation from Recommendation

True autonomy requires separating the estimation of what will happen from the recommendation of what we should do.

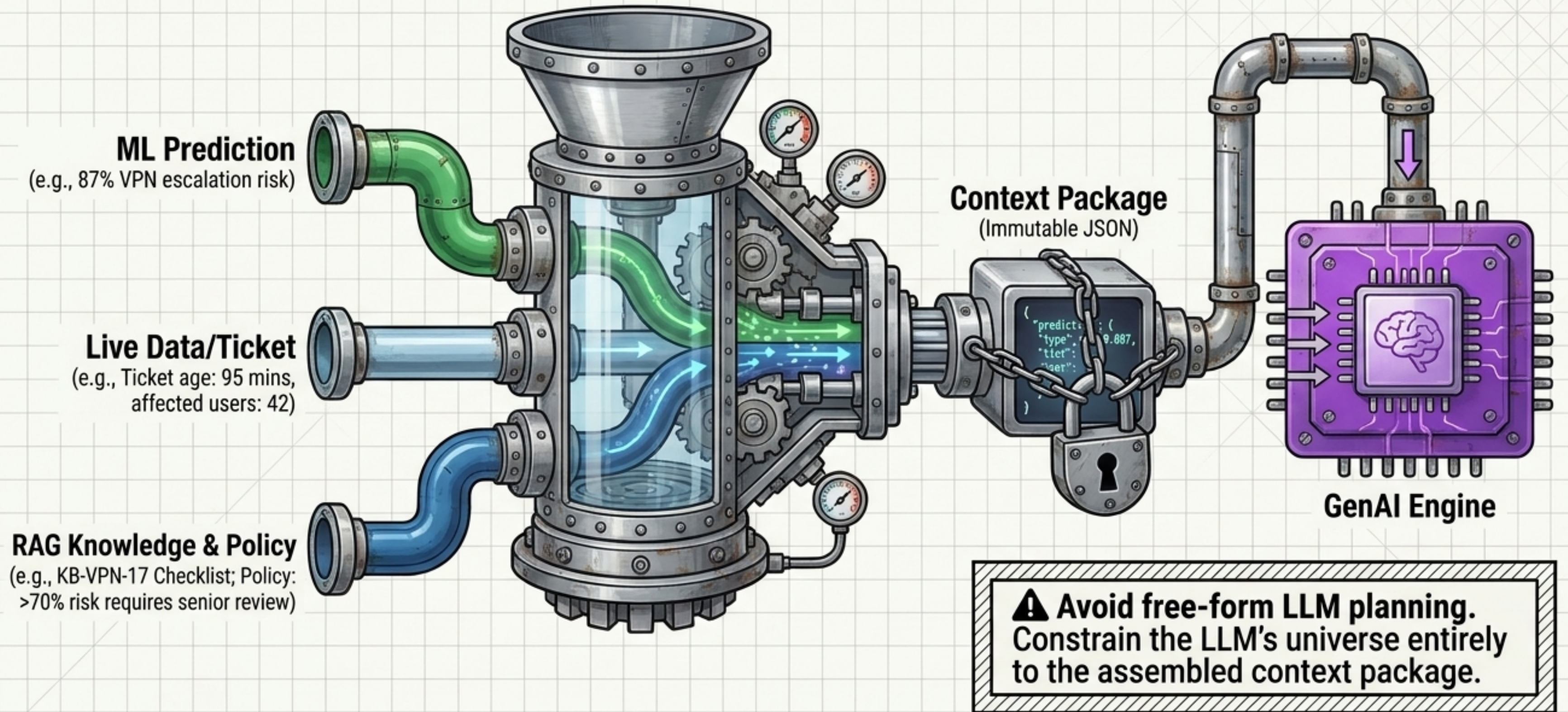


The End-to-End Production Blueprint

From signal to autonomous action: a closed-loop system of typed contracts.



The Context Mixer: Injecting Reality Before Generation



Trace: Predicting and Prescribing IT-Ticket Escalation

From signal to autonomous action:
a closed-loop system of typed contracts.

[Context - System]

Context Analyst Agent retrieves Policy POL-ESC-2 (threshold >0.70) and Knowledge KB-VPN-17.

[Prescribe - GenAI]

Recommendation Writer Agent synthesizes the package and drafts: Route to Network Operations; attach KB-VPN-17.

[Guardrail - Rules]

Deterministic validator confirms route_to_network is an allowlisted action string.

[Decide - Human]

Human UI presents the drafted narrative.
Human clicks APPROVE.

[Input - Signals]

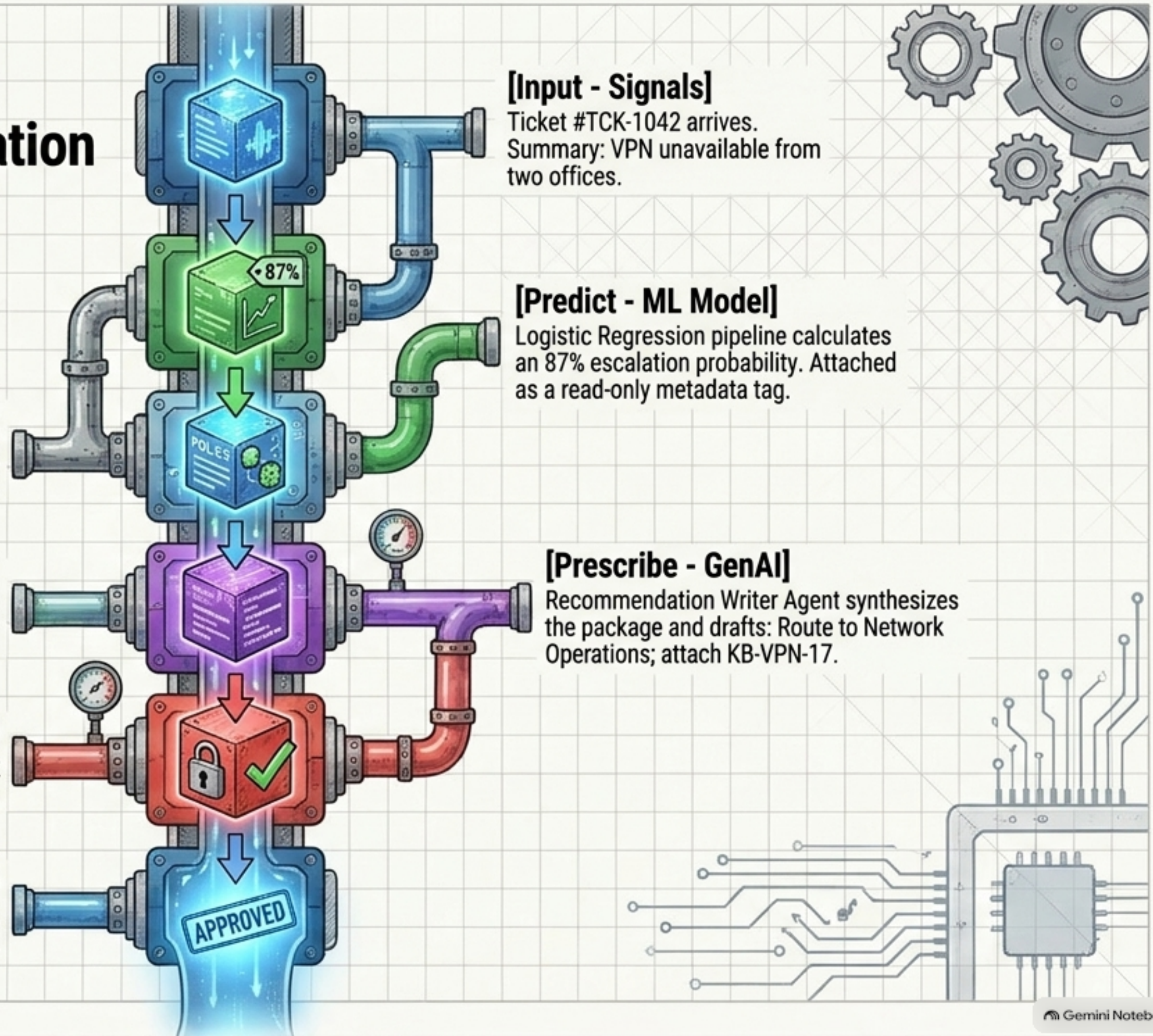
Ticket #TCK-1042 arrives.
Summary: VPN unavailable from two offices.

[Predict - ML Model]

Logistic Regression pipeline calculates an 87% escalation probability. Attached as a read-only metadata tag.

[Prescribe - GenAI]

Recommendation Writer Agent synthesizes the package and drafts: Route to Network Operations; attach KB-VPN-17.



Cross-Domain Modularity

The architectural boundary applies equally to predicting server outages or financial variances.

IT Operations

Predict: Logistic regression flags TCK-1042 with 87% escalation probability.

Prescribe: LLM retrieves KB-VPN-17 and proposes Route to Network Operations.

Decide: System requires human approval due to high probability.

Finance / Scenario

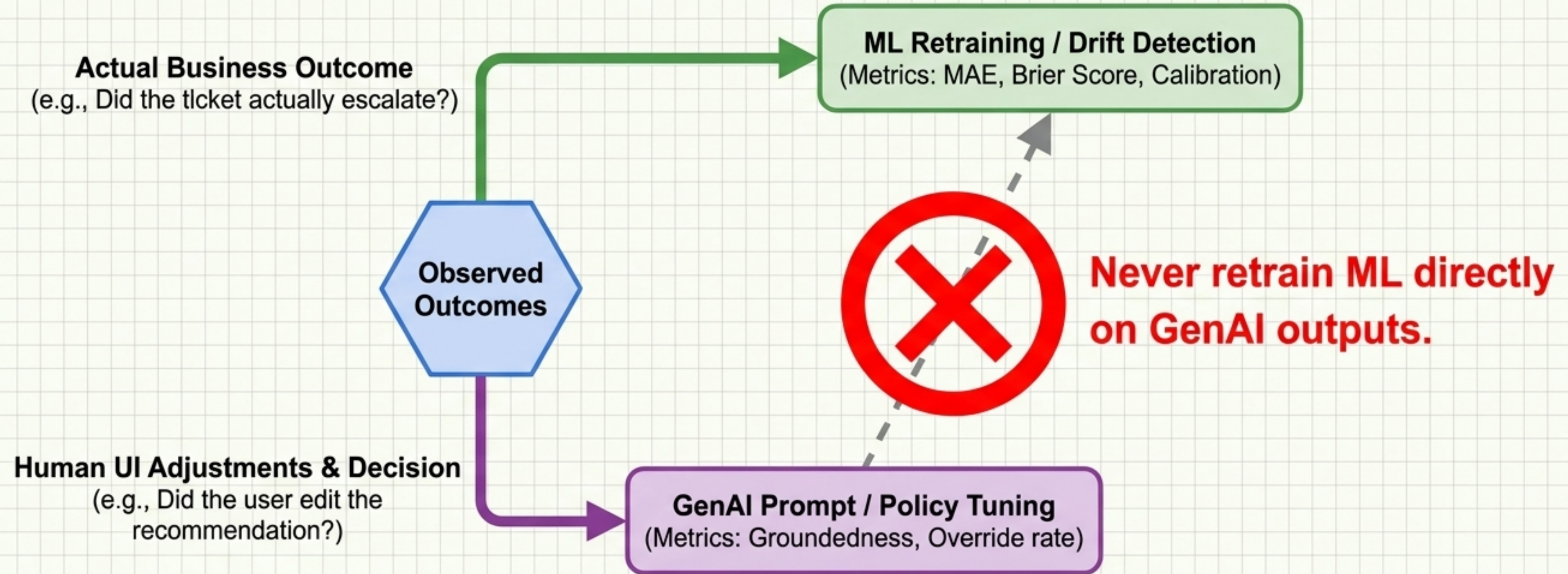
Predict: scikit-learn forecasts NOK 3.27M Q4 revenue (vs 3.50M budget).

Prescribe: LLM retrieves budget assumptions and explains the 6.6% variance.

Decide: GenAI uses a scenario tool to calculate a 5% downside drop; strictly barred from altering authoritative financial records.

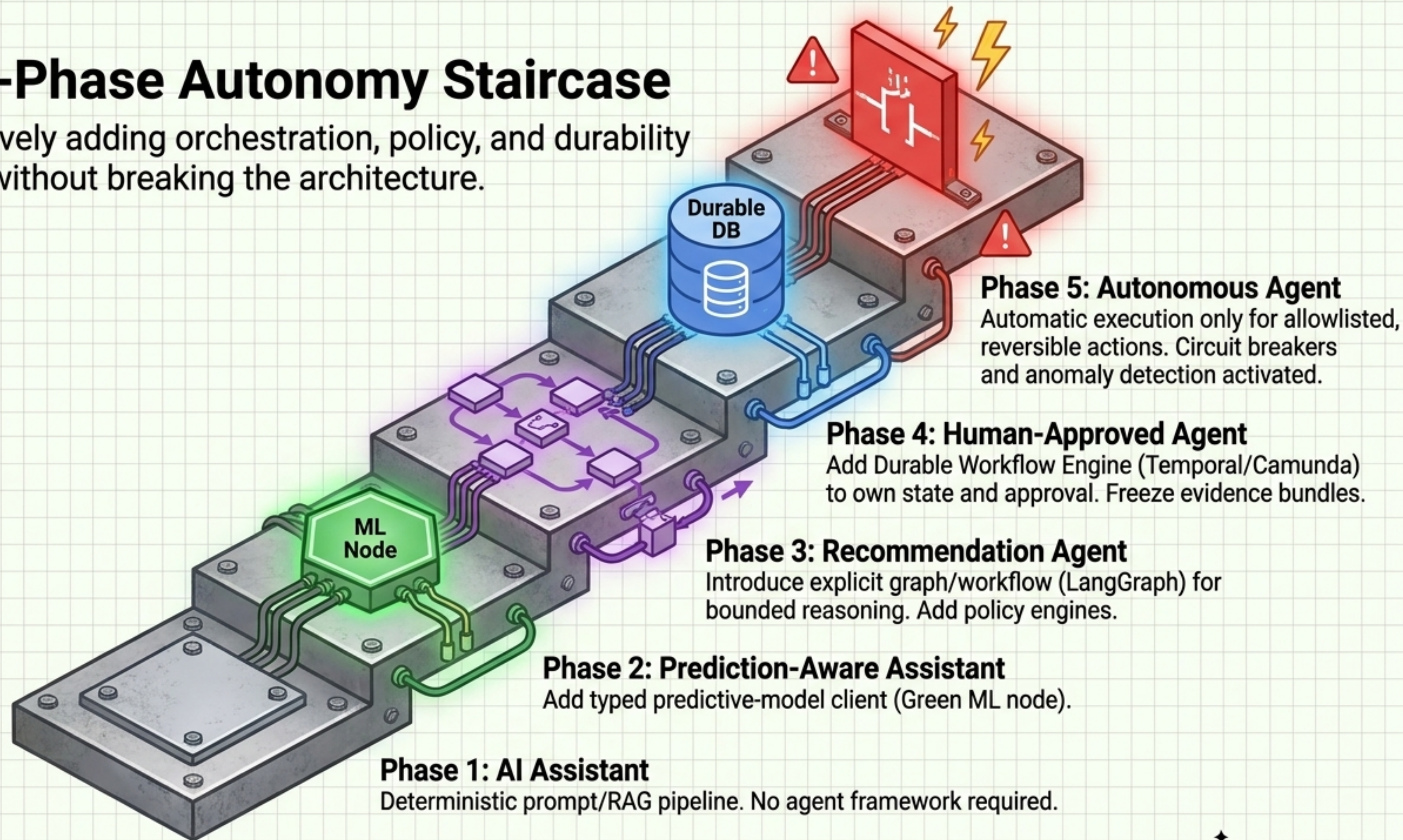
The Dual-Feedback Loop: Severing Self-Reinforcing Cycles

Evaluate predictive accuracy strictly separate from prescription utility.



The 5-Phase Autonomy Staircase

Progressively adding orchestration, policy, and durability controls without breaking the architecture.



Tooling Diagnostics: Agents vs. Workflow Engines

Never delegate high-impact durability or authorization solely to an agent loop.

Temporal / Camunda (Workflow Engines)	Durable state, retries, audit-oriented workflows.	Production default for Phases 4-5 (State & Execution). Not an LLM reasoning framework.
LangGraph (State Graphs)	Low-level persistence, deterministic + agentic nodes.	Strong cloud-neutral default for Phase 3 (Reasoning & Fact Gathering).
Microsoft Agent Framework / Google ADK	Typed tools, middleware, MCP.	Fast-moving platform alignments.

⚙ **Rule of Thumb:** Use **deterministic code** for known transformations, a workflow engine for known processes, an **optimization engine** for **constrained choices**, and an **LLM agent** only where the next step genuinely depends on interpreting ambiguous context. ⚙

The Golden Rules of the Prediction-to-Prescription Pipeline.

Rule of thumb: Use deterministic code for known transformations, a workflow engine for known processes, an optimization engine for constrained choices, and an LLM agent only for interpreting ambiguity.

